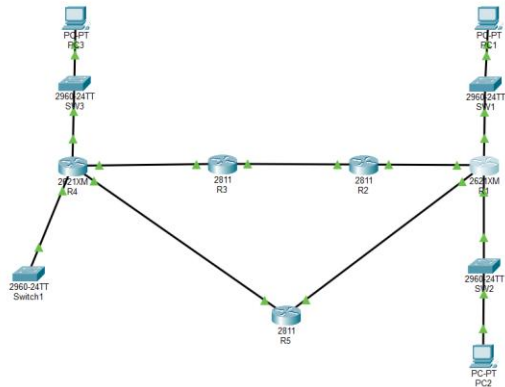


Practical week 7

Topology



Task 52

Display Name	R1	Display Name	R2	Display Name	R3	Display Name	R4	Display Name	R5
Hostname	R1	Hostname	R2	Hostname	R3	Hostname	R4	Hostname	R5

Task 53

IP Configuration	
IPv4 Address	10.0.2.1
Subnet Mask	255.255.255.0
IP Configuration	
IPv4 Address	10.0.3.1
Subnet Mask	255.255.255.0

Task 54

```
R1# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 2 subnets
C       10.0.2.0 is directly connected, FastEthernet1/0
C       10.0.3.0 is directly connected, FastEthernet1/1
```

Task 55

```
C:\> ping 10.0.2.10

Pinging 10.0.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\>tracert 10.0.2.10

Tracing route to 10.0.2.10 over a maximum of 30 hops:

  1  *          *          *          Request timed out.
  2  *          *          *          Request timed out.
  3  *          *          *          Request timed out.
  4  *          *          *          Request timed out.
  5  *          *          *          Request timed out.
  6  *          *          *          Request timed out.
  7  *          *          *          Request timed out.
  8  *          *          *          Request timed out.
  9  *          *          *          Request timed out.
 10 *          *          *          Request timed out.
```

Task 56

```
C:\> ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Task 57

IP Configuration	
IPv4 Address	10.0.0.2
Subnet Mask	255.255.255.0
R2	
IP Configuration	
IPv4 Address	10.1.1.2
Subnet Mask	255.255.255.0
R3	
IP Configuration	
IPv4 Address	10.1.1.1
Subnet Mask	255.255.255.0
R4	

Task 58

```

C:\> ping 10.1.2.1

Pinging 10.1.2.1 with 32 bytes of data:

Reply from 10.1.2.1: bytes=32 time<1ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255
Reply from 10.1.2.1: bytes=32 time=12ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.1.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

```

Task 59

```

R1(config)# ip route 10.0.0.0 255.255.255.0 10.1.2.1
R1(config)# 0.0.0.0 0.0.0.0 10.1.0.1
^
% Invalid input detected at '^' marker.

R1(config)# ip route 0.0.0.0 0.0.0.0 10.1.0.1
Enter configuration commands, one per line. End with
R2(config)# ip route 10.1.0.0 255.255.255.0 10.1.1.1
R2(config)# ip route 0.0.0.0 0.0.0.0 10.0.1.10

R3(config)# ip route 10.1.1.0 255.255.255.0 10.1.0.10
R3(config)# ip route 0.0.0.0 0.0.0.0 10.0.2.10

R4(config)# ip route 10.1.1.0 255.255.255.0 10.0.2.10
R4(config)# ip route 0.0.0.0 0.0.0.0 10.0.1.10

```

Task 60

```

Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

PC1

```

Pinging 10.0.1.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.1.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

PC2

```

Pinging 10.0.1.10 with 32 bytes of data:

Reply from 10.1.2.1: Destination host unreachable.
Reply from 10.1.2.1: Destination host unreachable.
Request timed out.
Reply from 10.1.2.1: Destination host unreachable.

Ping statistics for 10.0.1.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

PC3

Task 61

```

R1(config)# no ip route 10.0.0.0 255.255.255.0 10.1.2.1
R1(config)# no ip route 0.0.0.0 0.0.0.0 10.1.0.1

```

```
C:\> ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Task 62

```
R1(config)# ip route 10.0.0.0 255.255.255.0 10.1.2.10
```

Task 63

```
R1# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
    10.0.0.0/24 is subnetted, 4 subnets
C       10.0.0.0 is directly connected, FastEthernet0/0
C       10.0.1.0 is directly connected, FastEthernet0/1
C       10.0.2.0 is directly connected, FastEthernet1/0
C       10.0.3.0 is directly connected, FastEthernet1/1
```

```
C:\> ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Task 64

IP Configuration	
IPv4 Address	10.1.3.2
Subnet Mask	255.255.255.0

IP Configuration	
IPv4 Address	10.0.3.2
Subnet Mask	255.255.255.0

Task 65

```
C:\> ping 10.1.3.2

Pinging 10.1.3.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.3.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Task 66

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 4 subnets
C      10.0.0.0 is directly connected, FastEthernet0/0
C      10.0.1.0 is directly connected, FastEthernet0/1
C      10.0.2.0 is directly connected, FastEthernet1/0
C      10.0.3.0 is directly connected, FastEthernet1/1
```

Task 67

```
R5>enable
R5#traceroute
Protocol [ip]:
Target IP address: 10.0.1.10
Source address: 10.0.3.2
Numeric display [n]:
Timeout in seconds [3]:
Probe count [3]:
Minimum Time to Live [1]:
Maximum Time to Live [30]:
Type escape sequence to abort.
Tracing the route to 10.0.1.10

 1  *    *    *
 2  *    *    *
 3  *    *    *
 4  *    *    *
 5  *    *    *
 6  *    *    *
```

Task 68

IP Configuration	
IPv4 Address	203.0.113.1
Subnet Mask	255.255.255.0

Task 69

```
R4(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.2
```

Task 70

```
SW1(config)#port-channel load-balance ?
dst-ip      Dst IP Addr
dst-mac     Dst Mac Addr
src-dst-ip  Src XOR Dst IP Addr
src-dst-mac Src XOR Dst Mac Addr
src-ip      Src IP Addr
src-mac     Src Mac Addr
SW1(config)#port-channel load-balance src-dst-ip

SW2(config)#port-channel load-balance ?
dst-ip      Dst IP Addr
dst-mac     Dst Mac Addr
src-dst-ip  Src XOR Dst IP Addr
src-dst-mac Src XOR Dst Mac Addr
src-ip      Src IP Addr
src-mac     Src Mac Addr
SW2(config)#port-channel load-balance src-dst-ip
```

Task 71

Yes. All the interfaces are connected to route the addresses.

Task 72

PC1 is too far from PC3.

Task 73

It makes it simpler for the router to configure.